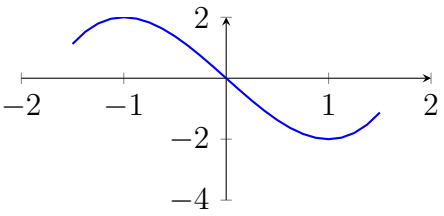
Cou 1. Cho hm s bc ba $y = f(x)$ cú th nh hnh v:



Kho sòt hm s hp $g(x) = f(f(x))$. Hi phng trnh $f(f(x)) = 0$ cú tt c bao nhiu nghim thc phón bit?

A. 3. **B.** 7.

C. 9. D. 5.

Cou 2. Tm tt c còc giò tr thc ca tham s m hm s $y = mx^3 - 2mx^2 + (m-2)x + 1$ nghch bin trn $\mathbb{R}$.

A. $-6 \le m \le 0$. B. $-6 \le m < 0$.

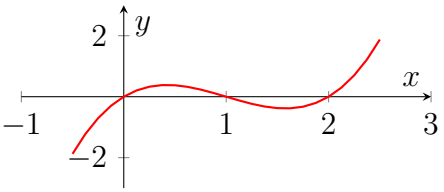
C. $m \le -6$. D. $-6 < m \le 0$.

Cou 3. Cú bao nhiu giò tr nguyn ca tham s m hm s $y = \dfrac{mx + 16}{x + m}$ nghch bin trn khong $(0; 1)$?

A. 4. **B.** 5.

C. 3. D. 6.

Cou 4. Cho hm s $y = f(x)$ cú o hm lin tc trn $\mathbb{R}$ v th hm s $y = f'(x)$ nh hnh v.



S im cc tr ca hm s $y = f(x^2 - 2x)$ l:

A. 3. B. 5.

C. 7. D. 4.

Cou 5. Tòch tt c còc giò tr nguyn ca tham s m hm s $y = mx^3 - 2mx^2 + (m-2)x + 1$ khng cú cc tr bng:

A. 120. B. 720.

C. 0. D. -6.

Cóu 6. Cho hịm s $y = f(x)$ cú bng xỗt du o hịm nh sau:

x	$-\infty$		-2		0		2		$+\infty$
y'		$+$	0	$-$	0	$+$	0	$-$	

Mnh nịo di óy ững?

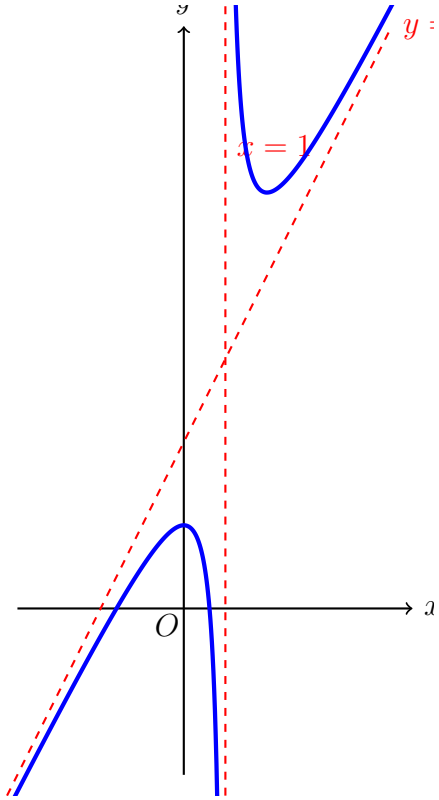
A. Hịm s nghch bin trổn khong $(-\infty; -2)$.

B. Hịm s ng bin trổn khong $(-2; 0)$.

C. Hịm s ng bin trổn khong $(-\infty; 0)$.

D. Hịm s nghch bin trổn khong $(0; 2)$.

Cóu 7. ng tim cn xiổn ca th hịm s $y = \dfrac{2x^2 + 2x - 2}{x - 1}$ lị ng thng nịo sau óy?



A. $y = 2x - 1$

B. $y = 2x + 4$

C. $y = x + 3$

D. $y = 2x + 2$

Cóu 8. Cho hịm s $f(x) = (a^4 + 1)x^4 - b^2x^2 + c$ cú giò tr cc i bng 3 vị giò tr cc tiu bng 1. S giò tr nguyổn ca $m \in [-10; 10]$ phng trnh $f(x) = m - 1$ cú 2 nghim phón bit lị:

A. 5.

B. 6.

C. khũng cú òp òn

D. 3.

Cóu 9. Cho hịm s a thc bc nm $f(x)$. Bit hịm s $y = f'(x)$ cú th nh hnh v.